Syllable token frequencies reveal a high cross-linguistic prevalence of the Sonority Sequencing Principle

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Abstract

The Sonority Sequencing Principle (SSP) is a phonotactic rule that constrains the order of phonemes inside a syllable based on their perceptual salience (sonority). According to the SSP, sonority increases from onset to nucleus and decreases toward the coda in well-formed syllables, whereas ill-formed syllables show a drop in sonority before the nucleus. Experimental evidence points to a preference for sonority-compliant syllables, even with materials not attested in the speakers' language or with very limited language experience (newborns), suggesting that the SSP modulates speech processing from early ages. The SSP is considered a linguistic universal that humans may be equipped with from birth, but whether SSP-following syllables are actually more common than ill-formed ones across languages is unclear. We addressed this issue by quantifying the prevalence of the SSP in a dataset of 34,878 syllables from 65 languages (13 macro-families), using syllabified phonological transcriptions while accounting for how often syllables are uttered in natural speech. SSP-following syllables were, on average, significantly more frequent than SSP-violating ones across languages, with an overall cross-linguistic conformity of 98.02%; most languages in our sample followed the SSP more than 90% of the time. Our findings suggest that SSP-following syllables are strongly preferred across languages and that the SSP is a cross-linguistically available constraint on syllabic structure. Notably, the syllabic structures most prevalent across languages are also preferred by learners from the earliest stages of development, consistent with the view that the structure of natural languages reflects the learning mechanisms humans bring to language acquisition.

Keywords: sonority, syllable, language acquisition, phonology, bayesian model

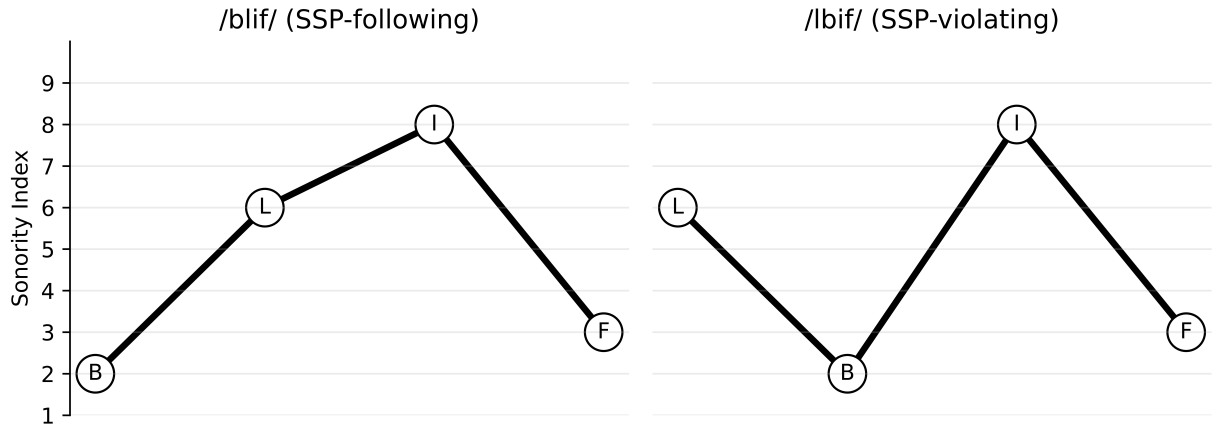
Syllable token frequencies reveal a high cross-linguistic prevalence of the Sonority Sequencing Principle

Sonority is a phonological feature that describes the perceptual salience of a given phoneme relative to neighbouring ones (Blevins, 1995; S. Parker, 2008; S. G. Parker, 2002; Selkirk, 1984). While the acoustic correlates of sonority are debated, some physical features such as intensity have been shown to correlate strongly with sonority indices in spoken languages (Fant et al., 2000; Jany et al., 2007; S. Parker, 2017; S. G. Parker, 2002; but see Albert & Nicenboim, 2022 for a different perspective). Phonemes can be hierarchically organised according to their sonority, with those at the lower end having lower sonority and those at the higher end having higher sonority (e.g., Obstruents < Nasals < Liquids < Glides < Vowels, Clements, 1990). This hierarchy is often represented as a numerical scale (e.g., 1 for lowest sonority, 9 for highest sonority, S. Parker, 2008). Based on this sonority hierarchy, some syllables are considered well-formed across languages while others are marked as ill-formed.

The Sonority Sequencing Principle (SSP) is a phonotactic rule that captures how sonority constrains the ordering of phonemes within a syllable. The SSP states that each syllable should display a unique peak of sonority at the nucleus, and that sonority should rise toward the nucleus and fall away from it (S. G. Parker, 2002; Selkirk, 1984). Syllables are classified as following or violating the SSP by examining the sequence of sonority values assigned to each of its phonemes Figure 1.

Figure 1

Schematic representation of the sonority hierarchy. The syllable /blif/ (left panel) has a sonority contour of 2-6-8-3 (8 being the vowel /i/, nucleus of the syllable) and is therefore considered well-formed, whereas /lbif/ (6-2-8-3, right panel) is considered ill-formed because the onset displays a sonority dip from the first to the second phoneme, violating the progressive increase in sonority that characterizes well-formed syllables.



Experimental evidence suggests that the SSP plays a pivotal role in processing spoken (Berent et al., 2007, 2011; Berent & Lennertz, 2010; Miozzo & Buchwald, 2013; S. Parker, 2017) and sign languages (Perlmutter, 1993; Sandler, 1993; Stone et al., 2018). Studies on newborns, infants, and adults show a consistent preference toward well-formed syllables (SSP-following) over ill-formed ones (SSP-violating), even when well-formed syllables are unattested in the participants' native language. For example, English-native adults prefer /bn/ (SSP-following) over /lf/ (SSP-violating) at syllable onset, although neither consonant cluster occurs in English (Berent et al., 2011; Berent & Lennertz, 2010). Few-day-old newborns show different brain haemodynamic responses when listening to SSP-following or violating syllables (e.g., *bran* vs. *lbif*) suggesting that precursors of the SSP are already present at birth (Gomez et al., 2014,

[CITATION REDACTED FOR ANONYMITY]). Such preferences for sonority-compliant syllables, even with unattested materials or very limited language experience (as in newborn infants) have been considered supporting evidence for the status of the SSP as a linguistic universal with which humans might be equipped from birth.

The universal status of the SSP is, however, still a matter of ongoing debate. Violations of the SSP are not uncommon across languages, with examples including fricative-stop consonant clusters at syllable onset (e.g., /ʃt/, as in the German word *Stein*) and their reverse counterpart at coda, stop-fricative clusters (e.g., /ks/, as in the English word *clicks*) (Kawasaki-Fukumori et al., 1997; Morelli et al., 2003). Languages that generally allow considerably complex consonant clusters at syllable onset tend to show SSP violations more frequently. This for instance the case of Polish, which allows for clusters of up to three consonants at syllable onset (e.g., *chrząszcz* /ˈxʂɔw̃ʂtʂ/ [beetle]) (Gussmann, 2007; Jarosz, 2017), Georgian (e.g., მწვრთნელი /mt͡sʼvrtʰneli/ [trainer]) (Crouch et al., 2023), or Russian (e.g., *дворянство* /dvoˈrʲɐnstvo/ [nobility]) (Halle, 1959).

A handful of studies have provided a quantitative approach toward the cross-linguistic prevalence of the SSP. Evidence of a preference for SSP-following consonant clusters across 43 languages was reported, with type frequencies of 97 distinct consonant clusters reflecting an overall 72% compliance with the SSP (Frisch, 2015). Type frequencies capture the number of distinct items (*types*, such as consonant clusters or syllables) in which a linguistic pattern (e.g., SSP-following structures) is present. When type frequencies were analyzed across 496 languages from 58 language families, it was found that 32% of the consonant clusters at word onsets violated the SSP, compared with 30% at word-final position (Yin et al., 2023). Both studies relied on type frequencies of attested consonant clusters in a sample of languages, counting the

number of consonant clusters that followed or violated the SSP in each of those languages, or simply establishing the presence or absence of SSP violations in them. While this approach provides useful information about languages' tolerance of SSP violations (or lack thereof), it does not provide a direct estimate of how often speakers of such languages encounter SSP violations in natural speech. This is because type frequency does not necessarily correspond on a one-to-one basis to its rate of occurrence in natural speech. Consider an imaginary language that allows for 20 consonant clusters in total, one of which is /st/. While /st/ represents a sizable portion of the language's phonotactical repertoire (5%), how frequently speakers encounter /st/ depends on at least two factors: the number of words containing /st/, and the frequency of each of those words. If /st/ occurs in only a few words or only in low-frequency words, its share in naturalistic speech would be lower than suggested by the language's phonological repertoire.

Another pitfall is that the phonological transcriptions used in most studies are drawn from lexical inventories and often lack syllable boundaries (Yin et al., 2023), making it difficult to draw reliable conclusions about the prevalence of the SSP in natural speech. The lack of syllabic boundaries results in an overepresentation of word-initial and word-final syllables, while word-medial consonant clusters remain unobserved, leaving a substantial portion of the syllabic structures uttered in natural speech unaccounted for.

In the present study, we aim to overcome these two pitfalls by providing a quantitative estimate of the cross-linguistic prevalence of the Sonority Sequencing Principle (SSP) that takes into account the frequency with which speakers encounter SSP-following and SSP-violating structures in natural language. To this end, we considered syllabic token frequency, which is a measure of how often a particular

syllable is uttered per million words in natural speech in a given language, and has been shown to be relevant in language processing and production tasks (Cholin & Levelt, 2009; Duchon et al., 2013; Perea & Carreiras, 1998). Additionally, to extract the syllabic repertoire of each language in the dataset, we used a large sample of syllabified phonological transcriptions that included word-medial syllables, which were not always represented in previous work. We then used a Bayesian modelling approach to estimate the prevalence of the SSP in a large dataset comprising 34,878 syllable types across 65 languages, yielding estimates of the proportion of SSP-following syllables in each language adjusted for the relative frequency of each syllable in natural speech.

Methods

Raw data, reproducible code, and project documentation are available in the Zenodo ([LINK REDACTED FOR ANONIMITY]) and GitHub ([LINK REDACTED FOR ANONIMITY]).

Data collection and preprocessing

Phonological transcriptions were scraped from word entries in the Wiktionary database. Wiktionary is an open-source, free database containing 10,050,751 lexical entries from 8,244 languages contributed by 4,463,976 users (as of June 10th, 2026), many of which include at least one phonological transcription for a given word. The wikipron Python package (version 2.2.0, Lee et al., 2020) was used to scrape all available phonological transcriptions of words for all 7,859 languages included in the Glottolog database (Nordhoff & Hammarström, 2011). This resulted in approximately 1.7 million phonological transcriptions across 900 languages. Languages with less than 100 words were excluded, resulting in a final dataset of 134,989 words from 75 languages.

Available lexical frequencies for each word in each language were extracted using the *wordfreq* Python package (Speer, 2022), which provides an interface for

retrieving lexical frequency data for words in over 40 languages. When lexical frequencies were not available for a particular language (e.g., the ISO 639-3 code did not match any language available in *wordfreq*), the closest equivalent language was used, if available. Equivalent languages were identified based on their distance from the original language according to the CLDR v38.11 standard[1]. The rationale for using equivalent languages to retrieve missing lexical frequencies was that users of close languages are likely to use similar words to refer to the same objects, and lexical frequencies were therefore expected to be reasonably comparable across equivalent languages. Words with missing lexical frequency scores were excluded from the dataset.

For the phonological transcription of each word, annotations about tonal features were stripped, then each phonological transcription was segmented into syllable candidates. When possible, syllable boundary markers were used, either explicit (".") or implicit ("ˈ" for primary stress, "ˌ" for secondary stress). In the absence of syllable markers, the word was assumed to be monosyllabic. Segments of each syllable candidate were then classified as consonants (C), vowels (V), or semi-vowels (v). Individual vowels, consecutive strings of vowels and semi-vowels (e.g., V, VV, Vv, VVv) were classified as vocalic clusters. Syllable candidates with more than one vocalic cluster (e.g., CVvCV, VCV) were flagged as multisyllabic and excluded from the dataset. This method allowed us to extract word-medial syllables when possible. Duplicated syllables within each language were removed (e.g., if the syllable /ka/ appeared in the

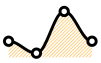
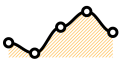
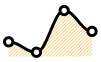
[1]The CLDR system assigns a numeric value to each pair of language codes based on whether both codes are assigned to identical languages with different names (e.g., Norwegian Bokmål to Norwegian, score 0), minor regional differences (score 1-3, e.g., Norwegian Bokmål to Norwegian), and languages with significant but unproblematic regional differences (score 10-24, e.g., Afrikaans to Dutch, Wu Chinese to Mandarin Chinese), etc.

Spanish words /ˈka.sa/ and /kaˈpo/, it was counted only once in the Spanish dataset). This procedure yielded one syllabic repertoire for each language.

To evaluate whether each extracted syllable followed or violated the SSP, a sonority score was assigned to each phoneme on a scale from 1 to 9, using the *panphon* Python package (version 0.22.2, Mortensen et al., 2016). Syllables with more than one peak (i.e., exhibiting a local minimum in the sonority contour) were classified as SSP-violating, and as SSP-following otherwise. Following this rule, syllables with a flat sonority contour (e.g., English word *sphinxs* /ˈsfɪŋks/ in which the first two phonemes bear identical sonority index value, 3-3-8-5-1-3) were classified as SSP-following. Whether consecutive segments with same sonority index comply or not with the SSP is a matter of debate (e.g., Tropf, 1987; Berent et al., 2007; Eckman, 1987; Santolin et al., 2023), and this reason we report complementary analyses in which flat sonority contours were classified as SSP violating, showing equivalent results (see Appendix C).

Table 1

Examples of SSP-following and SSP-violating syllables in the dataset. Sonority index contours are shown for each syllable numerically and graphically. For example, the syllable /ˈskul/ in Assamese shows a violation at onset, as sonority drops from /s/ (sonority index 3) to /k/ (sonority index 1) before rising again at nucleus /u/ (sonority index 8).

Language	Syllable	Original word	Sonority index	
SSP-following				
Modern Greek	/bɾos/	λάμπροσ | /ˈlam.bɾos/	2-6-9-3	
Finnish	/koht/	suhtkoht | /ˈsuht̩koht/	1-9-6-1	
Hindi	/njäːɾ/	न्यार | /njäːr/	5-7-9-6	
Russian	/font/	фонт | /font/	3-9-5-1	
SSP-violating				
Modern Greek	/ftis/	συγγραφτείσ | /siŋ.ɣraˈftis/	3-1-8-3	
Finnish	/s̪trøm/	ångström | /ˈo̰ŋ.s̪trøm/	3-1-6-9-5	
Hindi	/st̪ẽn/	स्तनपायियो | /st̪ẽn.päː.ji.joː/	3-1-9-5	
Russian	/fpeɬ/	вполглаза | /fpeɬˈgɬazə/	3-1-9-6	

To calculate the frequency of each syllable, we summed the lexical frequency (counts per million words) of all words in the dataset which contained that syllable (see Duchon et al., 2013 for a similar approach). The rationale behind this approach is that, whenever a word is uttered, all of its syllables are necessarily uttered as well. Therefore, summing the lexical frequencies of all words containing a given syllable yields an

estimate of that syllable's frequency in natural speech. The final dataset comprised 34,878 syllables from 65 languages from 13 distinct language macro-families.

Data analysis

We used a multilevel Bayesian regression model to estimate the probability of each syllable in the dataset following (or violating) the SSP, considering the language it belonged to and its syllabic frequency. Bayesian models estimate a probability distribution for each parameter in the model, known as the *posterior* distribution. This is obtained by combining prior knowledge about the parameter (specified as a *prior* distribution based on domain knowledge) with the observed data via the Bayes rule. The posterior distribution provides inferential information about how likely each parameter is to be equal or close to zero. To estimate the posterior distribution of the model we used Hamiltonian Monte Carlo (HMC), an algorithm that explores the posterior distribution iteratively, as implemented in *PyMC3* (version 6.2.0, Patil et al., 2010) with four chains of 2,000 iterations, 1,000 of which were warm-up iterations.

We modelled the response variable y_{ij} (1 if the syllable was SSP-following, 0 otherwise) using a Bernoulli distribution with probability parameter θ_{ij}. We specified a linear regression model for θ_{ij} which included fixed regression coefficients for the intercept (α) and the slope of syllabic frequency (β), and random effects u_j and w_j for each fixed coefficient, respectively. We specified weakly informative priors for all parameters in the model (see Equation 1 in Appendix A).

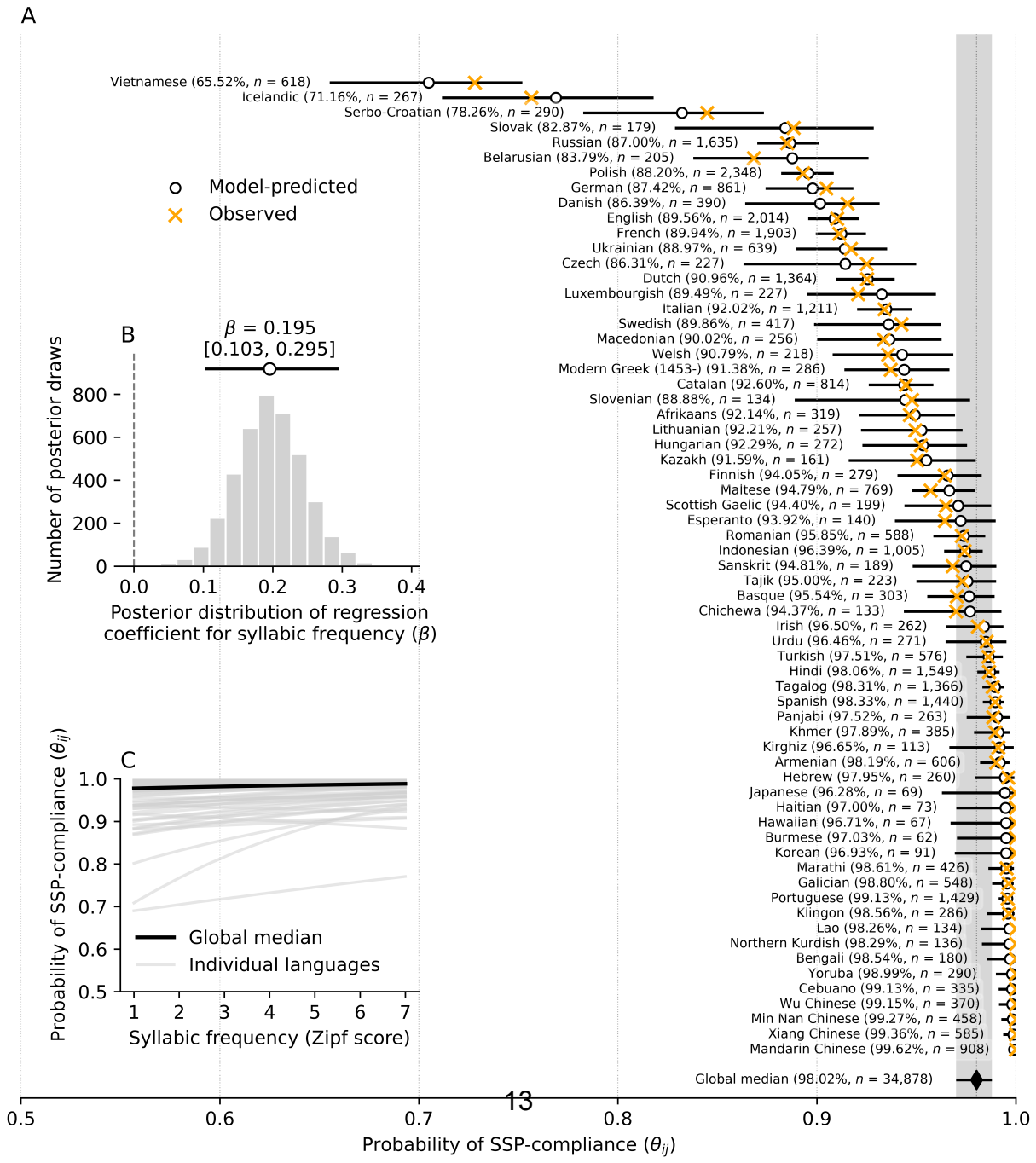
We report the median and 95% credible interval (CrI) of the posterior distribution for the parameters of interest. The 95% CrI can be interpreted as the interval that has 95% probability of containing the true value of the parameter (Kruschke & Liddell, 2018). Parameters whose 95% CrI excluded zero were concluded to be reliably different from zero.

Results

The global proportion of SSP-following syllables across languages was θ = 0.980 (95% CrI = [0.970, 0.988]), and the prevalence of the SSP for each language ranged from 70.498% (Vietnamese, 95% CrI = [65.516%, 75.197%]) to 99.908% (Mandarin Chinese, 95% CrI = [99.622%, 99.989%]) (see Figure 2). Only 3.08% of the languages in the sample showed less than 80% of SSP prevalence, 12.31% languages showed less than 90% of SSP prevalence, and 35.38% languages showed less than 95% of SSP prevalence. Syllabic frequency showed a positive association with the probability of SSP compliance (β = 0.195, 95% CrI = [0.103, 0.295]), indicating that SSP-following syllables were, on average, more frequent than SSP-violating syllables.

Figure 2

A) Observed and model-predicted SSP prevalence (median and 95% CrI). B) Posterior distribution of the regression coefficient for syllabic frequency (β). More positive or negative values show a more positive or negative association between this predictor and SSP compliance (logit scale). C) Model-predicted of SSP compliance as a function of syllabic frequency. Growing lines indicate a positive association between both variables.



Discussion

Previous experimental studies had pointed to the Sonority Sequencing Principle (SSP) as a universal constraint defining the structure of syllables, key units in speech processing (e.g., Berent & Lennertz, 2010; Berent et al., 2011; Gomez et al., 2014, [CITATION REDACTED FOR ANONIMITY]). However, it remained unclear to what extent well-formed syllables according to the SSP are indeed more common than ill-formed ones across languages. The present study addressed this issue by quantifying the prevalence of the SSP across a diverse set of languages, using syllabified phonological transcriptions that include all syllables composing words (not just word-initial or final ones) while also accounting for how often syllables are uttered in natural speech. Results revealed that SSP-following syllables are, on average, significantly more frequent than SSP-violating ones across the languages in our dataset, with an overall cross-linguistic conformity of 98.022%. In addition, the majority of languages in our sample followed the SSP more than 90% of the time.

A key novel aspect of the current study compared to prior work is the use of syllabic token frequencies (instead of type frequencies) which more closely reflect how often speakers actually encounter SSP-following or SSP-violating syllables in everyday speech. If SSP-following syllables occur more often than SSP-violating ones, estimates of SSP prevalence likely underestimate their true value when syllabic token frequencies are not incorporated into the analysis. This may partially explain the higher prevalence of SSP compliance found here compared to other studies (e.g., 72% of SSP-following consonant clusters in 47 languages vs. 98.022% in this study Frisch, 2015; see also Watson, 2002; Al Tamimi & Al Shboul, 2013; Gishri, 2009; Yin et al., 2023). Consistent with this possibility, we found a positive, non-zero association between syllabic token frequency and the probability of SSP compliance, suggesting that SSP-following

syllables tend to be uttered more frequently in natural speech. Taken together, our results suggest that most languages in our dataset comply with the SSP, albeit to varying degrees, and that violations of the SSP are exceptional for the majority of languages.

Complementarily, we also found that SSP violations, while uncommon, are attested in most languages in our dataset (see Yin et al., 2023 for a similar pattern of results). There are a few methodological considerations to keep in mind when interpreting these findings. First, syllables included in our dataset were not randomly sampled. Instead, each syllable was included based on the availability of at least one word containing that syllable in the corresponding language in Wiktionary, which may have introduced some degree of selection bias into our dataset (see Yin et al., 2023 for a similar pattern of results). Second, while the present study includes multiple languages from a range of macro-families, the number of observations available for some languages is reduced, especially when comparing Indo-European languages with those from other families. This unequal data availability across languages is, however, partially accounted for by the partial pooling of random effects in multilevel models (see Gelman et al., 2007 for a detailed explanation).

Our findings can be interpreted within the Optimality Theory (OT) framework (Frisch, 2015; Prince & Smolensky, 2004), according to which linguistic universals are not hard, binary rules but rather general biases towards certain type of structures that most languages favour. Within this framework, exceptions to the SSP can be understood as the result of the SSP interacting with other phonotactic rules that speakers of certain languages weigh more strongly. Under OT, sonority constraints on syllabic structure remain useful for generating predictions about the well-formedness of

syllables, despite the cross-linguistic variability in the degree of compliance with this principle (Dell & Elmedlaoui, 1985).

Previous experimental work had capitalized on cross-linguistic differences in SSP compliance to investigate the role of sonority constraints on speech perception. For instance, Russian native adults correctly identified SSP-violating syllables (e.g., *lbif*) as monosyllabic, whereas English native adults misperceived them as a disyllabic (e.g., *lebif*) splitting the ill-formed consonant cluster /lb/ in two syllables (/le/ and /bif/, Berent et al., 2007). These results were attributed to Russian speakers' more frequent exposure to SSP-violating syllables compared to English speakers (Halle, 1959). Our results suggest that Russian still shows an overall higher prevalence of SSP-following structures despite its lower compliance with the SSP compared to other languages (87.45% vs. 90.879% in English, for example). Similar results were found in other Slavic languages such as Polish (88.658%), Ukrainian (91.400%), and Czech (91.428%), in which SSP violations are traditionally known to be more frequent than in other language families. A possible explanation for the discrepancy between experimental work and our corpus results is that, although SSP compliance rates are high in both Russian and English, Russian speakers have more frequent exposure to SSP-violating syllables than English speakers. Even a relatively small difference in exposure to SSP-violating structures may be sufficient to shape speakers' perception of these structures, leading to the behavioral differences observed in experimental studies. Firther research is needed to determine whether language-wide estimates of SSP prevalence, such as those presented here, fully capture the relevant mechanisms that modulate the role of sonority in speech perception. Further research is needed to determine whether language-wide estimates of SSP prevalence, such as those presented here, fully

capture the relevant mechanisms that modulate the role of sonority in speech perception.

Overall, the present study updates prior estimates of cross-linguistic compliance with the SSP, showing 1) a substantially higher presence of SSP-following syllables than SSP-violating syllables across many languages, and that 2) the SSP-following syllables occur more frequently in everyday speech than the SSP-violating ones. It has been proposed that the learning mechanisms underlying language acquisition have shaped the structure of natural languages over time, such that their organization reflects the very abilities humans use to learn them (Chater & Christiansen, 2010; Christiansen & Chater, 2008, 2016; Saffran, 2001, 2002). Our findings are consistent with this view: the syllabic structures most prevalent across language families (SSP-following) are also those that human learners show a preference for from the earliest stages of language development.

References

Al Tamimi, Y. A., & Al Shboul, Y. (2013). Is the phonotactics of the Arabic complex coda sonority-based? *Journal of King Saud University-Languages and Translation*, *25*(1), 21–33.

Albert, A., & Nicenboim, B. (2022). Modeling Sonority in Terms of Pitch Intelligibility With the Nucleus Attraction Principle. *Cognitive Science*, *46*(7), e13161. https://doi.org/10.1111/cogs.13161

Berent, I., Harder, K., & Lennertz, T. (2011). Phonological Universals in Early Childhood: Evidence from Sonority Restrictions. *Language Acquisition*, *18*(4), 281–293. https://doi.org/10.1080/10489223.2011.580676

Berent, I., & Lennertz, T. (2010). Universal constraints on the sound structure of language: Phonological or acoustic? *Journal of Experimental Psychology:*

Human Perception and Performance, *36*(1), 212–223. https://doi.org/10.1037/a0017638

Berent, I., Steriade, D., Lennertz, T., & Vaknin, V. (2007). What we know about what we have never heard: Evidence from perceptual illusions. *Cognition*, *104*(3), 591–630. https://doi.org/10.1016/j.cognition.2006.05.015

Blevins, J. (1995). The Syllable in Phonological Theory. In J. Goldsmith (Ed.), *Handbook of phonological theory* (Vol. 1, pp. 206–244). Blackwell.

Chater, N., & Christiansen, M. H. (2010). Language acquisition meets language evolution. *Cognitive Science*, *34*(7), 1131–1157.

Cholin, J., & Levelt, W. J. (2009). Effects of syllable preparation and syllable frequency in speech production: Further evidence for syllabic units at a post-lexical level. *Language and Cognitive Processes*, *24*(5), 662–684.

Christiansen, M. H., & Chater, N. (2008). Brains, genes, and language evolution: A new synthesis. *BEHAV BRAIN SCI*, *31*(5), 537–558.

Christiansen, M. H., & Chater, N. (2016). The now-or-never bottleneck: A fundamental constraint on language. *Behavioral and Brain Sciences*, *39*.

Clements, G. N. (1990). The role of the sonority cycle in core syllabification. *Papers in Laboratory Phonology*, 283–333. https://doi.org/10.1017/cbo9780511627736.017

Crouch, C., Katsika, A., & Chitoran, I. (2023). Sonority sequencing and its relationship to articulatory timing in Georgian. *Journal of the International Phonetic Association*, *53*(3), 1049–1072.

Dell, F., & Elmedlaoui, M. (1985). Syllabic consonants and syllabification in imdlawn tashlhiyt berber. *Journal ofAfrican Languages and Linguistics*, *7*, 105–130.

Duchon, A., Perea, M., Sebastián-Gallés, N., Martí, A., & Carreiras, M. (2013). EsPal: One-stop shopping for spanish word properties. *Behavior Research Methods*, *45*(4), 1246–1258.

Eckman, F. R. (1987). The reduction of word-final consonant clusters in interlanguage. *Sound Patterns in Second Language Acquisition*, 143–162.

Fant, G., Kruckenberg, A., & Liljencrants, J. (2000). Acoustic-phonetic analysis of prominence in swedish. In *Intonation: Analysis, modelling and technology* (pp. 55–86). Springer.

Frisch, S. A. (2015). A preliminary investigation of quantitative patterns in sonority sequencing. *Italian Journal of Linguistics*, *27*(1), 9–27.

Gelman, A., Hill, J., et al. (2007). *Data analysis using regression and multilevel/ hierarchical models* (Vol. 1). Cambridge university press Cambridge.

Gishri, M. (2009). *The role of PROMI e CE in the acquisition of medial codas in hebrew* [PhD thesis]. Tel Aviv University.

Gomez, D., Berent, I., Benavides-Varela, S., Bion, R., Cattarossi, L., Nespor, M., & Mehler, J. (2014). Language universals at birth. *Proceedings of the National Academy of Sciences*, *111*(16), 5837–5841.

Gussmann, E. (2007). *The phonology of polish*. OUP Oxford.

Halle, M. (1959). *The sound pattern of russian: A linguistic and acoustical investigation*. Mouton. https://books.google.es/books?id=hqTzsRp7OgkC

Jany, C., Gordon, M., Nash, C. M., & Takara, N. (2007). How universal is the sonority hierarchy?: A cross-linguistic acoustic study. *Proceedings of the ICPhS*, 1401–1404.

Jarosz, G. (2017). Defying the stimulus: Acquisition of complex onsets in polish. *Phonology*, *34*(2), 269–298.

Kawasaki-Fukumori, H., Ohala, J., Eliasson, S., & Jahr, E. (1997). Alternatives to the sonority hierarchy for explaining segmental sequential constraints. *Language and Its Ecology: Essays in Memory of Einar Haugen*, *100*, 343.

Kruschke, J. K., & Liddell, T. M. (2018). The Bayesian New Statistics: Hypothesis testing, estimation, meta-analysis, and power analysis from a Bayesian perspective. *Psychonomic Bulletin & Review*, *25*(1), 178–206.

Lee, J. L., Ashby, L. F., Garza, M. E., Lee-Sikka, Y., Miller, S., Wong, A., McCarthy, A. D., & Gorman, K. (2020). Massively multilingual pronunciation modeling with WikiPron. *Proceedings of the Twelfth Language Resources and Evaluation Conference*, 4223–4228.

Miozzo, M., & Buchwald, A. (2013). On the nature of sonority in spoken word production: Evidence from neuropsychology. *Cognition*, *128*(3), 287–301.

Morelli, F., Féry, C., & Vijver, R. van de. (2003). *The syllable in optimality theory*.

Mortensen, D. R., Littell, P., Bharadwaj, A., Goyal, K., Dyer, C., & Levin, L. (2016). Panphon: A resource for mapping IPA segments to articulatory feature vectors. *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, 3475–3484.

Nordhoff, S., & Hammarström, H. (2011). Glottolog/Langdoc: Defining dialects, languages, and language families as collections of resources. In T. Kauppinen, L. C. Pouchard, & K. Keßler (Eds.), *Proceedings of the first international workshop on linked science 2011 (LISC2011), bonn, germany, october 24, 2011* (pp. 1–7). CEUR Workshop Proceedings.

Parker, S. (2008). Sound level protrusions as physical correlates of sonority. *Journal of Phonetics*, *36*(1), 55–90. https://doi.org/10.1016/j.wocn.2007.09.003

Parker, S. (2017). Sounding out Sonority. *Language and Linguistics Compass*, *11*(9), e12248. https://doi.org/10.1111/lnc3.12248

Parker, S. G. (2002). *Quantifying the sonority hierarchy*. University of Massachusetts Amherst.

Patil, A., Huard, D., & Fonnesbeck, C. J. (2010). PyMC: Bayesian stochastic modelling in python. *Journal of Statistical Software*, *35*, 1–81.

Perea, M., & Carreiras, M. (1998). Effects of syllable frequency and syllable neighborhood frequency in visual word recognition. *Journal of Experimental Psychology: Human Perception and Performance*, *24*(1), 134.

Perlmutter, D. M. (1993). Sonority and syllable structure in american sign language. In *Current issues in ASL phonology* (pp. 227–261). Elsevier.

Prince, A., & Smolensky, P. (2004). Syllable Structure Typology: The CV Theory. In *Optimality Theory* (pp. 105–118). John Wiley & Sons, Ltd. https://doi.org/10.1002/9780470759400.ch6

Saffran, J. R. (2001). Words in a sea of sounds: The output of infant statistical learning. *Cognition*, *81*(2), 149–169.

Saffran, J. R. (2002). Constraints on statistical language learning. *Journal of Memory and Language*, *47*(1), 172–196.

Sandler, W. (1993). A sonority cycle in american sign language. *Phonology*, *10*(2), 243–279.

Santolin, C., Crespo-Bojorque, P., Sebastian-Galles, N., & Toro, J. M. (2023). Sensitivity to the sonority sequencing principle in rats (Rattus norvegicus). *Scientific Reports*, *13*(1), 17036. https://doi.org/10.1038/s41598-023-44081-y

Selkirk, E. (1984). On the major class features and syllable theory. *Language Sound Structure*.

Speer, R. (2022). *Rspeer/wordfreq: V3.0*. Zenodo. https://doi.org/10.5281/zenodo.7199437

Stone, A., Petitto, L.-A., & Bosworth, R. (2018). Visual sonority modulates infants' attraction to sign language. *Language Learning and Development*, *14*(2), 130–148.

Tropf, H. (1987). Sonority as a variability factor in second language phonology. *Sound Patterns in Second Language Acquisition*, 173–191.

Watson, J. C. (2002). *The phonology and morphology of arabic*. OUP Oxford.

Yin, R., van de Weijer, J., & Round, E. R. (2023). Frequent violation of the sonority sequencing principle in hundreds of languages: How often and by which sequences? *Linguistic Typology*, *27*(2), 381–403.

Appendix A

Model structure

Likelihood:

$$y_{ij} \sim \text{Bernoulli } (\frac{\theta_{ij}}{1-\theta_{ij}})$$

$$\theta_{ij} = \alpha + u_j + (\beta + w_j) \cdot \text{Frequency}_{ij}$$

Prior:

$$\begin{aligned} \alpha, \beta &\sim \text{Normal } (0, 5) \\ u_j, w_j &\sim \text{Normal } (0, \sigma_{u,w}) \\ \sigma_{u,w} &\sim \text{HalfNormal } (1) \end{aligned} \tag{1}$$

Where:

i indexes each observation and j indexes the language it belongs to

y_{ij} indexes whether the syllable follows (1) or violates (0) the SSP

u_j, w_j are language-specific adjustments to α and β

Appendix B

Number of SSP-following and SSP-violating syllables per language

Table B1

Number of available words and syllables available in the dataset, and number and proportion of SSP-following syllables for each language in the dataset.

ISO-639-3	Words	Syllables	SSP+	%	ISO-639-3	Words	Syllables	SSP+	%
kmr	141	136	136	100.0%	nya	108	133	129	97.0%
ceb	417	335	335	100.0%	san	145	189	183	96.8%
wuu	1,481	370	370	100.0%	gla	512	199	192	96.5%
haw	128	67	67	100.0%	epo	1,451	140	135	96.4%
mya	443	62	62	100.0%	fin	8,619	279	269	96.4%
jpn	3,651	69	69	100.0%	mlt	871	769	736	95.7%
ben	871	180	180	100.0%	hun	4,778	272	259	95.2%
cmn	2,198	908	908	100.0%	kaz	214	161	153	95.0%
lao	124	134	134	100.0%	lit	1,497	257	244	94.9%
hsn	877	585	585	100.0%	slv	646	134	127	94.8%
nan	1,157	458	458	100.0%	afr	330	319	302	94.7%
hat	133	73	73	100.0%	cat	1,852	814	769	94.5%
yor	350	290	290	100.0%	swe	551	417	393	94.2%
kor	2,618	91	91	100.0%	ell	637	286	268	93.7%
tlh	306	286	285	99.7%	cym	511	218	204	93.6%
glg	905	548	546	99.6%	ita	4,858	1,211	1,131	93.4%
heb	257	260	259	99.6%	mkd	2,802	256	239	93.4%
por	4,986	1,429	1,423	99.6%	nld	1,934	1,364	1,262	92.5%
mar	418	426	424	99.5%	ces	2,410	227	210	92.5%
kir	105	113	112	99.1%	ltz	369	227	209	92.1%
hye	1,444	606	600	99.0%	ukr	2,199	639	586	91.7%
khm	647	385	381	99.0%	dan	724	390	357	91.5%
spa	11,578	1,440	1,425	99.0%	fra	6,398	1,903	1,734	91.1%
pan	333	263	260	98.9%	eng	4,665	2,014	1,833	91.0%
tgl	3,045	1,366	1,350	98.8%	deu	2,146	861	779	90.5%
hin	3,254	1,549	1,528	98.6%	pol	5,634	2,348	2,097	89.3%
tur	527	576	568	98.6%	slk	2,337	179	159	88.8%
urd	223	271	267	98.5%	rus	17,447	1,635	1,447	88.5%
gle	593	262	257	98.1%	bel	611	205	178	86.8%
ind	3,208	1,005	979	97.4%	hbs	3,299	290	245	84.5%
tgk	267	223	217	97.3%	isl	1,049	267	202	75.7%
ron	987	588	572	97.3%	vie	1,060	618	450	72.8%
eus	1,328	303	294	97.0%	-	-	-	-	-

Appendix C

Results after classifying flat sonority contours as SSP-violating

Figure C1

A) Observed and model-predicted SSP prevalence (median and 95% CrI). B) Posterior distribution of the regression coefficient for syllabic frequency (β). More positive or negative values show a more positive or negative association between this predictor and SSP compliance (logit scale). C) Model-predicted of SSP compliance as a function of syllabic frequency. Growing lines indicate a positive association between both variables.

